

AR for the Masses: Attina, the Low-Cost Accessible Headset for Inclusive Learning

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Figure 1: A side view of the Attina headset with a mounted smartphone (left). A breakdown of Attina's individual components and assembly tools (middle). A user interacting with the AR content projected by Attina (right).

ABSTRACT

Current commercial Augmented Reality (AR) headsets are expensive and limited in availability, posing challenges for researchers and designers in resource-constrained settings. To address this, we introduce Attina, an open-source, low-cost optical see-through AR headset using a smartphone as its display. Overcoming limitations of previous work, the system features stereoscopic rendering for enhanced depth perception and a 3-DoF controller for virtual interactions. We discuss Attina's design and outline opportunities for a usability study to evaluate its user experience.

CCS CONCEPTS

• **Computing methodologies** → **Mixed / augmented reality**; *Virtual reality*; • **Human-centered computing** → **Mixed / augmented reality**; *Virtual reality*; *Ubiquitous and mobile computing systems and tools*; *Accessibility*.

KEYWORDS

Augmented Reality; Education; Inclusivity; Accessibility

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1 INTRODUCTION AND RELATED WORK

Despite AR technology's enormous potential and a market capitalization projected to reach \$97.76 billion by 2028 [7], the high cost and limited availability of commercial AR headsets like Microsoft HoloLens 2¹ (\$3,500) and the Magic Leap 2² (\$3,299) hinder access for many potential users. There is a pressing need for accessible AR solutions in developing countries [1], given the proven benefits of visual immersion for memory retention and recall in education [9]. Providing such solutions is crucial for unlocking AR's potential in fostering educational advancements in these settings. Attina addresses this gap by offering a low-cost, DIY AR headset utilizing 3D-printed parts, a combiner, and a smartphone, with a total cost of 47 USD, making AR technology more attainable for a wider demographic.

Previous research has aimed to make AR more accessible by creating smartphone-based headsets. Amer and Perez [2] developed a low-cost headset (12 USD), that used concave lenses obtained from welding goggles as eyepieces, and the headset in its entirety was made of foam board. It was primarily built for Virtual Reality, but the smartphone's camera could be exploited for a video pass-through AR experience. However, it required manual adjustments, lacked adequate quality of experience, and did not include a controller for virtual content interaction.

¹<https://www.microsoft.com/en-us/hololens/buy>, retrieved July 27, 2023

²<https://www.magicleap.com/where-to-buy>, retrieved July 27, 2023

Brun et al. [4] introduced a cardboard-based DIY headset, CAR-TON, featuring an SDK for developer access and two sample apps. The headset employed a transparent plastic sheet as the combiner, with a mirror angled inwards and facing the smartphone's camera such that the camera can see what the users' eyes see, enabling context-aware experiences. However, it relied on head gestures for interaction, lacked a controller, and only functioned as a heads-up display, limiting the scope of AR experiences possible.

Another headset is by De Angeli and O'Neill [5], which also exploits the Pepper's Ghost approach and is targeted towards museum AR experiences. It featured a periscope system with a mirror angled towards the smartphone's camera for tracking the user's view and displaying virtual content accurately. However, their study found that blue and red texts were mostly invisible under certain lighting conditions, and the headset did not enable user interaction with the content. Meanwhile, the Zapbox [15] by Zappar is a commercial 6-DoF headset developed for use with smartphones and allows for interactions through controllers, although it is priced at 80 USD, works on video pass-through, and so far only supports iOS smartphones. The OpenAR 2.0 [12] and the Project Northstar [11] are also promising headsets but incorporate expensive modules such as micro displays [13] and 6-DoF sensors that drive up the costs.

In summary, efforts have been made to increase AR accessibility using low-cost, smartphone-based headsets. Nevertheless, these devices exhibit certain constraints that impede the full realization of AR experiences. Attina offers an affordable, comprehensive, and well-documented open-source solution, addressing the shortcomings of prior work, and making it easier to deploy immersive learning within resource-limited environments.

2 THE ATTINA APPROACH

In response to the accessibility challenges posed by the exorbitant price points of commercially available augmented reality head-mounted displays, we have developed a modular and cost-effective AR headset which leverages the computational and display capabilities of a ubiquitous smartphone.

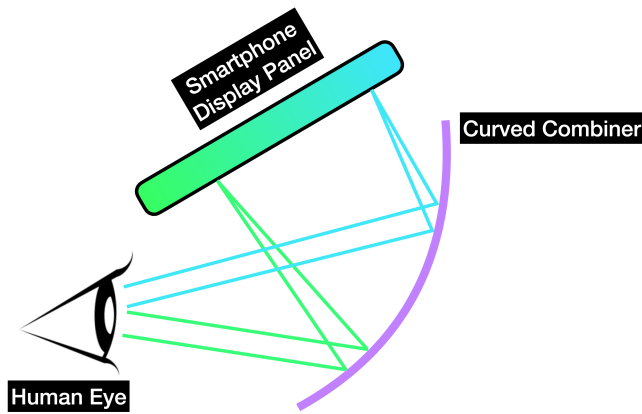


Figure 2: Visual representation depicting the direction of light within Attina.

Design and Implementation. The headset design features key components (Figure 1), including a 3D-printed enclosure for replicability. A reflective combiner, functioning as a beam splitter (an optical medium to split a beam in two), enables seamless virtual content overlay on the user's field of view using the Pepper's Ghost effect³. The smartphone's display is angled at 40° towards the combiner, reflecting light from the display into the user's eyes, as seen in Figure 2. A head strap with a rotary dial, acquired from a welding gear harness, ensures easy adjustments and a secure fit. To optimize positioning on the head and better weight distribution, a frontal incision is incorporated, enabling the utilization of an overhead velcro strap. Additionally, cushions are positioned at high-pressure regions to enhance user comfort. The AR application is rendered on a smartphone housed within the headset, in a slot compatible with most contemporary models, with a maximum thickness of 10mm. The smartphone is held securely by two tension-based supporters. The combiner is secured by an adjustable fastener, allowing users to switch to real-world view, thus granting eye relief.

AR Experience. We utilized Unity v2020.3.12f1⁴ to develop the software, ensuring cross-platform compatibility and supporting both iOS and Android smartphones. To improve depth perception, we employed a stereoscopic rendering approach, rendering the scene twice on each half of the smartphone screen by placing two virtual orthographic cameras at a 63.5mm inter-camera distance, angled inwards to converge at a single user viewpoint. This distance, the average Interpupillary Distance [10], can be customized for user preference to enhance stereoscopic acuity. We also flipped the cameras' projection matrix to counteract the mirror effect caused by the combiner reflection. Our headset also incorporates head-tracking capabilities using built-in accelerometer and gyroscope sensors in the smartphone, enabling 3-DoF tracking for interactive experiences. Moreover, the curved UI enhances depth, immersion, and engagement by surrounding the user's field of view, as well as addressing eyestrain [3]. A limitation of our work is the slight distortion in peripheral vision caused by the curved combiner. However, this issue can be mitigated with software enhancements, which we plan to address in a future update. Lastly, it should be noted that the visual quality of the AR experience is subject to the resolution of the smartphone screen, as the combiner directly reflects it.

Wireless Controller. Interaction with virtual content is vital for immersive AR and diverse experiences. To facilitate this, we designed a 3-DoF controller for point-and-click interactions, consisting of a NodeMCU⁵ for wireless data transmission, MPU 6050 Inertial Measurement Unit⁶ for 3-DoF tracking, a power switch, a tactile button for selections, a battery, and a charging controller designed to prevent overcharging (Figure 3). The controller's motion data is wirelessly transmitted to the AR application via UDP packets through a socket connection on the same network.

To use the laser pointer in AR, users must calibrate the controller by holding it in the desired position and pressing the tactile button for three seconds. During calibration, the system calculates the quaternion angle between a calibration vector, positioned at the

³<https://www.scienceworld.ca/resource/peppers-ghost-hologram-illusion/>, retrieved July 27, 2023

⁴<https://unity.com>, retrieved July 27, 2023

⁵https://www.nodemcu.com/index_en.html, retrieved July 27, 2023

⁶<https://www.arduino.cc/reference/en/libraries/mpu6050/>, retrieved July 27, 2023

center of the virtual content, and the direction vector of the laser pointer. Based on this calculation, a transformation is applied to ensure that the pointer accurately aligns with the virtual content. This simple and efficient calibration process enables user-friendly and precise interactions.

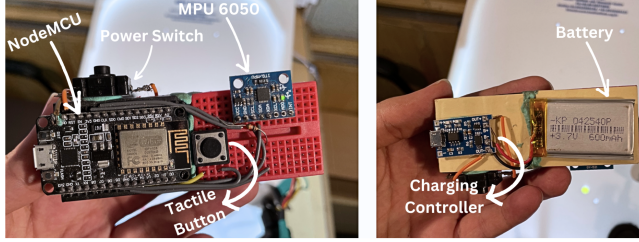


Figure 3: Annotated images of the wireless controller, depicting the top view (left), and the bottom view (right).

3 CONCLUSION

In conclusion, Attina is an open-source AR headset utilizing 3D-printed components, a combiner, and a smartphone to provide an immersive 3-DoF AR experience for advancing learning in resource-constrained settings. The bill of materials, STL files for 3D printing, assembly instructions, and software can be found at https://github.com/rajshekarreddy97/Attina_AugmentedReality_HMD. We encourage researchers and developers to build upon this framework and contribute to the advancement (e.g. adding 6-DoF capabilities) of affordable and accessible AR technology.

Limitations and Future Work. Limitations include the restricted field of view resulting from distortions in the peripheral vision, and the brightness constraints of the smartphone's display limiting indoor use. Despite efforts to incorporate readily available, off-the-shelf components, we cannot ensure universal accessibility of its modules (e.g., smartphones and 3D printers). Nonetheless, we intend to soon include various sources and online vendors for 3D printing in our documentation. Future work could focus on evaluating Attina's performance through user studies. Specifically, to assess the headset's usability, comfort, and overall user experience in diverse application domains by comparing it with other low-cost and high-end AR headsets. Future work could also explore incorporating additional sensors, such as the Intel Realsense T265 [8] for 6-DoF tracking or Leap Motion Controller [14] for hand tracking. Additionally, investigating OpenXR [6] compatibility could simplify integration with diverse AR applications.

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